

The Hermite-Minkowski Theorem

Give the ring of integers $\mathcal{O}_K$ in a number field K , we can embed it as a lattice as follows.

The finite $\mathbb{R}$ -algebra $A = \mathbb{R} \otimes_{\mathbb{Z}} \mathcal{O}_K$ is isomorphic to $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$.

The $\mathbb{Z}$ -module $\mathcal{O}_K$ has a basis $S = \{\alpha_1, \dots, \alpha_n\}$ where n is the degree of the field extension K of $\mathbb{Q}$.

Thus, we have a map $\mathcal{O}_K \rightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$ given by

$$\alpha_i \mapsto \mathbf{a}_i = \left(x_i^{(1)}, \dots, x_i^{(r_1)}, z_i^{(1)}, \overline{z_i^{(1)}}, \dots, z_i^{(r_2)}, \overline{z_i^{(r_2)}} \right)$$

whose image is a lattice in $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$. Note that if $M_S(\alpha_i)$ is the $n \times n$ matrix describing multiplication by α_i on K , then $\mathbf{a}_i$ is the tuple of its real and complex eigenvalues.

If A is the $n \times n$ matrix $[\mathbf{a}_1, \dots, \mathbf{a}_n]$, then $B = A^t A$ is the matrix whose (i, j) -th entry is

$$\mathbf{a}_i \cdot \mathbf{a}_j = x_i^{(1)} x_j^{(1)} + \dots + x_i^{(r_1)} x_j^{(r_1)} + z_i^{(1)} z_j^{(1)} + \overline{z_i^{(1)}} \overline{z_j^{(1)}} + \dots + z_i^{(r_2)} z_j^{(r_2)} + \overline{z_i^{(r_2)}} \overline{z_j^{(r_2)}}$$

By the above description of $\mathbf{a}_i$, we see that this is $\text{Tr}_{K/\mathbb{Q}}(\alpha_i \cdot \alpha_j)$. It follows that $\det(B) = \det(A)^2$ is the discriminant D_K of the ring $\mathcal{O}_K$.

The volume of the lattice $\mathcal{O}_K$ is typically taken by writing $\mathbb{C} = \mathbb{R} \times \mathbb{R}$ by taking real and imaginary parts, so as to write $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$ as $\mathbb{R}^n$. In other words, instead of the above generators of the lattice, we should use

$$\tilde{\mathbf{a}}_i = \left(x_i^{(1)}, \dots, x_i^{(r_1)}, \text{Re}(z_i^{(1)}), \text{Im}(z_i^{(1)}), \dots, \text{Re}(z_i^{(r_2)}), \text{Im}(z_i^{(r_2)}) \right)$$

This change of basis has determinant 2^{-r_2} .

In summary, the volume of the lattice $\mathcal{O}_K$ in $\mathbb{R}^n$ is given by $2^{-r_2} |D_K|^{1/2}$.

Assume that $r_2 \geq 1$ (if necessary replace K by $K(\sqrt{-1})$) and consider the symmetric convex domain

$$\Omega_C = \left\{ \left(x^{(1)}, \dots, x^{(r_1)}, \text{Re}(z^{(1)}), \text{Im}(z^{(1)}), \dots, \text{Re}(z^{(r_2)}), \text{Im}(z^{(r_2)}) \right) : \right. \\ \left. |x^{(i)}| < 1 \text{ for } i \in [1, r_1]; |z^{(j)}| < 1 \text{ for } j \in [1, r_2 - 1]; \right. \\ \left. |\text{Re}(z^{(r_2)})| < 1; |\text{Im}(z^{(r_2)})| < C \right\}$$

This domain has volume $4C \cdot 2^{r_1} \pi^{r_2-1}$. Thus, if

$$4C \cdot 2^{r_1} \pi^{r_2-1} > 2^n \cdot 2^{-r_2} |D_K|^{1/2}$$

there is a non-zero element α in $\Omega_C \cap \mathcal{O}_K$. We can take $C \geq (2/\pi)^{r_2-1} |D_K|^{1/2}$ for this to hold.

Since α is a non-zero algebraic integer we must have $|\text{Nm}_{K/\mathbb{Q}}(\alpha)| \geq 1$. Writing

$$\alpha = \left(x^{(1)}, \dots, x^{(r_1)}, \text{Re}(z^{(1)}), \text{Im}(z^{(1)}), \dots, \text{Re}(z^{(r_2)}), \text{Im}(z^{(r_2)}) \right)$$

as a point of $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$ this means

$$\prod_{i=1}^{r_1} |x^{(i)}| \cdot \prod_{j=1}^{r_2} |z^{(j)}|^2 \geq 1$$

Since we are given that this is a point in Ω_C , we have

$$|x^{(i)}| < 1 \text{ for } i \in [1, r_1] \text{ and } |z^{(j)}| < 1 \text{ for } j \in [1, r_2 - 1]$$

Thus, $|z^{(r_2)}| > 1$. Moreover, since $|\text{Re}(z^{(r_2)})| < 1$ we see that $|\text{Im}(z^{(r_2)})| > 1$.

This means that:

- $z^{(r_2)} \neq x^{(i)}$ for $i \in [1, r_1]$,
- $z^{(r_2)} \neq z^{(j)}$ for $j \in [1, r_2 - 1]$,
- $z^{(r_2)} \neq \overline{z^{(j)}}$ for $j \in [1, r_2 - 1]$,
- $z^{(r_2)} \neq \overline{z^{(r_2)}}$.

In other words, $z^{(r_2)}$ is *different* from *all* the other eigenvalues of $M_S(\alpha)$. Since the characteristic polynomial $M_S(\alpha)$ is a *power* of its minimal polynomial, this means that the characteristic polynomial of $M_S(\alpha)$ is *equal* to its minimal polynomial. Hence, we see that $K = \mathbb{Q}(\alpha)$.

If $|D_K|^{1/2} < (\pi/2)^{r_2-1}C$, then $K = \mathbb{Q}(\alpha)$ for α an algebraic integer whose image in $\mathbb{R}^n$ lies in Ω_C .

If α is an algebraic integer whose image in $\mathbb{R}^n$ lies in Ω_C , then the roots of $M_S(\alpha)$ are bounded. It follows that the coefficients of the characteristic polynomial of α are bounded. Since these coefficients are integers, there are at most finitely many such polynomials. It follows that there are at most finitely many α lying in Ω_C . Since α *determines* K , we obtain the following result.

There are at most finitely many fields K that are degree n extensions of $\mathbb{Q}$ with $|D_K|$ bounded by a fixed constant.